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TREND ANALYSIS AND PREDICTION OF MAXIMUM TEMPERATURE IN SOUTHERN THAILAND

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Abstract. *This study analyzes long-term trends in daily maximum temperatures across 21 stations in Southern Thailand and evaluates predictive models, including multiple linear regression (MLR), random forest, XGBoost, support vector regression (SVR), and artificial neural networks (ANN). Temperature trends show primary peaks in April-May, secondary peaks in July-August, and the lowest temperatures in December-January. A significant warming trend was observed at most stations, with Lanta Island and Narathiwat showing the highest increase (∼0.6°C per decade). Predictive models used four variables, including relative humidity, wind speed, wind direction, and rainfall. XGBoost outperformed other models at 13 stations, indicating its strength in capturing complex relationships. SVR performed best at six stations, while MLR was optimal at two, though with lower R^2 values. The second-best model analysis showed SVR as a strong competitor, ranking second at 12 stations. ANN was not among the top models, suggesting it requires more data for better generalization. Statistical testing confirmed XGBoost significantly outperformed ANN, while SVR, RF, and MLR showed comparable performance. These results highlight the effectiveness of tree-based and kernel-based methods for temperature forecasting in Southern Thailand.*

Keywords: Temperature trends, Machine learning, XGBoost, SVR, Southern Thailand

1. **Introduction.** People around the world are increasingly concerned about temperature changes for various reasons, including their impact on extreme weather [1,2], agriculture [3,4], and human health [5,6]. Research has revealed that the combination of drought and heat stress significantly impacts yield by decreasing the harvest index, shortening the life cycle of crops, and altering seed number, size, and composition [7]. A study in Southern China reported a significant relationship between drought/flood intensity and sugarcane climatic yield [8]. An analysis of heat-related fatalities in the Census of Fatal Occupational Injuries from 1992 to 2016 found a correlation between rising summer temperatures and increased heat-related death rates [9]. Similarly, another study using data from the North Carolina Disease Event Tracking and Epidemiologic Collection Tool examined emergency department visits for heat-related illness (HRI). The findings indicate that HRI rates rise significantly on days when daily maximum temperatures are unusually high, ranging from 31 to 38°C [10].

Numerous researchers have focused on the direction of maximum temperature variation [11,12] and examined the relationships between daily maximum temperature and other parameters [13-15]. For instance, the health implications of rising temperatures are equally alarming. In central Arizona, heat-related mortality increases exponentially at temperatures below the seasonal median, indicating that even moderate heat can pose

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significant risks [6,16]. Similarly, abnormally high daily maximum temperatures have been linked to a sharp rise in heat-related illnesses [6]. These findings underscore the urgent need to understand and predict temperature variations, particularly in regions vulnerable to extreme heat.

However, in Thailand, research on temperature patterns, trends, and predictions remains limited. While several studies have analyzed temperature variations in the northern [17,18] and central regions, including Bangkok [19], and some have explored spatial temperature patterns in Southern Thailand [20-22]. In fact, an investigation of average monthly maximum temperature in Thailand from 1986 to 2015 showed that over the last 10 years, the average maximum surface temperature in Southern Thailand had a warming trend of 0.068°C [18]. However, a comprehensive investigation into patterns, long-term trends, and predictive modeling of maximum temperatures in Southern Thailand is still lacking. As daily maximum temperature provides insights into extreme heat exposure, which is critical for understanding heat stress, heatwaves, and health risks. Additionally, given the increasing concerns about temperature changes and their potential consequences, there is a critical need for a systematic assessment of temperature patterns and trends and effective predictive models for this region.

Therefore, this study has two primary objectives. First, it aims to analyze seasonal patterns and long-term trends in daily maximum temperatures in Southern Thailand. Second, it seeks to identify the most suitable model to predict these temperatures. In particular, the seasonal patterns were investigated by the natural cubic spline method, the long-term trend was examined by simple linear regression, and the predictive models under consideration include multiple linear regression (MLR), random forest (RF), XGBoost, support vector regression (SVR), and artificial neural networks (ANN).

2. Materials and Methods.

2.1. Data and research framework. The data in this study were obtained from the Thai Meteorological Department, including five variables, namely, maximum temperature, relative humidity, wind speed, wind direction, and rainfall. They all are daily observations starting from January 1st, 2008, to December 31st, 2022 (5749 days), covering 21 stations of 12 provinces in Southern Thailand. The geological location of each station was illustrated in Figure 1.

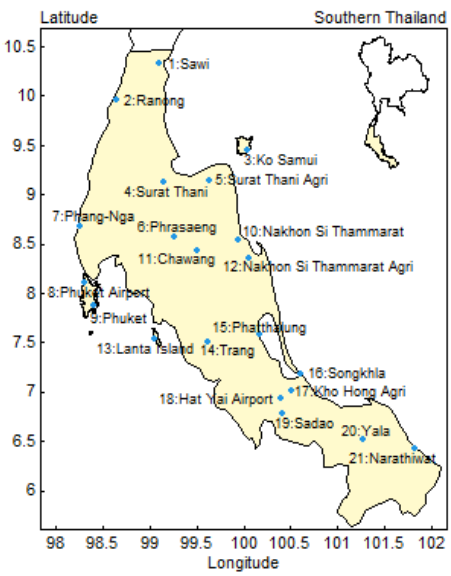


FIGURE 1. Geological map location of 21 MET stations in Southern Thailand

The brief distribution of the data was shown as the deviation about their means for daily maximum temperature, relative humidity, wind speed, and wind direction. However, since the rainfall data were very skewed, the 70th, and 90th percentiles of rainfall were provided instead (Table 1). The majority of stations have missing values that are less than 1%, with the exception of Chawang, which has 10.26%. The relative humidity data was quite complete. Most stations maintain very high relative humidity, with median values typically ranging from 75% to 85%.

Table 1. The mean±standard deviation (SD) of daily maximum temperature, relative humidity, wind speed, and wind direction, along with the 70th, and 90th percentiles of daily rainfall

Station	Mean±SD				Rainfall	
	TMax	R.Humidity	W.Speed	W.Direction	P75	P90
1. Chawang	33.31±2.35	83.52±6.14	8.43±3.77	169.56±89.74	8.00	21.14
2. Hat Yai Airport	32.84±2.17	81.01±6.52	15.50±5.25	154.07±105.63	5.40	17.80
3. Kho Hong Agri.	32.67±2.09	79.93±5.98	12.27±4.49	148.84±106.07	7.60	25.32
4. Ko Samui	31.30±2.07	81.5±5.57	14.17±5.45	171.65±93.07	4.70	19.04
5. Lanta Island	31.90±1.65	80.94±6.71	15.64±6.61	185.61±95.07	7.60	24.85
6. Nakhon Si Thammarat	32.71±2.24	82.64±5.66	13.44±3.95	148.25±95.64	6.60	24.20
7. Nakhon Si Tham. Agri.	32.47±2.27	85.28±4.91	8.72±4.42	143.44±98.53	7.90	27.25
8. Narathiwat	32.07±1.90	80.50±5.05	11.16±2.64	107.68±84.23	8.60	30.63
9. Phang-Nga	31.84±1.56	83.43±6.49	12.89±4.97	220.96±90.34	12.80	36.17
10. Phatthalung	32.37±2.16	82.59±5.76	11.73±5.46	174.41±92.63	6.50	24.20
11. Phrasaeng	33.15±2.19	83.53±5.69	11.18±3.09	178.01±101.6	6.50	17.45
12. Phuket	33.11±1.80	76.99±6.76	13.45±3.48	186.75±103.21	9.00	25.00
13. Phuket Airport	32.04±1.53	79.40±5.86	12.48±5.27	199.16±96.49	10.30	29.11
14. Ranong	31.98±2.18	79.27±7.81	13.71±4.56	184.58±95.38	16.90	45.00
15. Sadao	33.13±2.03	82.42±5.33	14.49±3.40	162.59±100.41	3.80	14.56
16. Sawi	32.14±2.19	81.05±4.59	5.59±3.87	130.77±99.2	6.40	20.74
17. Songkhla	32.00±1.94	78.04±5.56	10.05±2.53	136.28±92.85	5.90	22.43
18. Surat Thani	33.03±2.46	83.40±5.62	14.14±4.14	145.45±114.04	4.60	15.80
19. Surat Thani Agri.	32.45±2.16	83.11±5.84	8.18±2.92	175.34±91.04	6.05	21.00
20. Trang	33.07±2.18	81.21±7.38	14.42±3.73	161.53±101.05	8.40	23.70
21. Yala	33.16±2.31	81.27±5.92	14.59±4.63	161.93±94.85	7.60	25.91

Wind speed data from all stations exhibit missing values not exceeding 7%. Numerous stations see severe wind velocities beyond 100 knots, but standard extreme speeds typically range from 30 to 50 knots. The wind direction exhibits a comparable number of missing information, barely surpassing 9%. Wind directions encompass the entire 360-degree spectrum. The ultimate variable is precipitation. Rainfall data exhibit a maximum missing value proportion of 28%. The distribution pattern indicates a tropical monsoon climate, marked by numerous dry days or days with low precipitation, revealing minimal values at the 75th and 90th percentiles.

In accordance with the research framework shown in Figure 2, five meteorological variables – maximum temperature, relative humidity, wind speed, wind direction, and rainfall – were obtained from the Thai Meteorological Department. The original data were provided in a wide format, where columns represent dates within a month, and rows represented months and years. After reshaping, some unlikely outliers were set as NA, such as wind speeds exceeding 220 knots. Missing values were then imputed using the predictive mean matching (PMM) method.

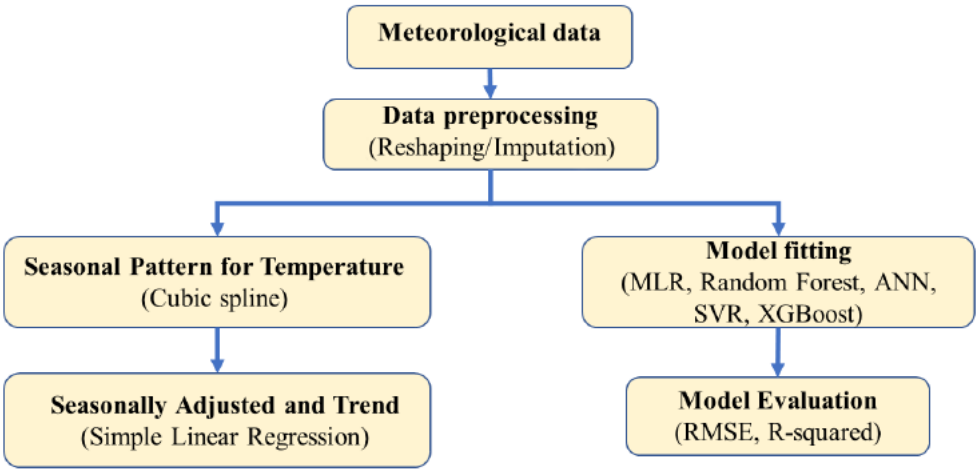


FIGURE 2. Research framework of the study

Following data preprocessing, two parallel analytical approaches were conducted. The first approach focused on analyzing seasonal temperature patterns using cubic spline techniques, followed by seasonally adjusted trend analysis through simple linear regression. The second approach involved the application of various machine learning models, including multiple linear regression, random forest, extreme gradient boosting, support vector regression, and artificial neural networks.

The dataset was partitioned into training (70%) and testing (30%) subsets. Model training was performed on the training subset, while the performance of the predictive models was assessed on the testing subset using two standard evaluation metrics: root mean square error (RMSE) and R-squared.

2.2. Methodology.

2.2.1. *Models for seasonal patterns and trends investigation.* To analyze the underlying patterns of maximum temperature, a natural cubic spline function with eight knots was constructed to smooth the average maximum temperature for each day of the year. Cubic splines offer flexibility across the entire data range, allowing for precise modeling of complex patterns. Furthermore, the use of natural cubic splines incorporates additional constraints, setting the second derivative to zero at the boundaries. This approach aligns with the assumption that seasonal patterns flatten or transition smoothly at the data edges, resulting in more stable and realistic extrapolations.

The seasonally adjusted maximum temperature was derived by removing the seasonal patterns from the original time series and adding the overall average. This adjustment ensured that the resulting seasonally adjusted temperature remained approximately within the same range as the original values. Subsequently, a simple linear regression model was applied to the seasonally adjusted maximum temperature to examine its temporal changes over the past 15 years (2008-2022).

2.2.2. *Predictive models.* Five predictive models – MLR, RF, XGBoost, SVR, and ANN – were employed to examine relative humidity, wind speed, wind direction, and rainfall. These models, widely used in weather data analysis, offer distinct advantages: MLR provides simplicity and insights into variable relationships, while tree-based methods (RF, XGBoost) excel with mixed data types and complex, non-linear relationships. Non-tree-based methods (SVR, ANN), though potentially requiring more effort for tuning, are effective for capturing intricate patterns. Each model was selected for its unique strengths in addressing the study's objectives. Firstly, the equation for MLR is given by

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \varepsilon,$$

where in this case, y is the maximum temperature, β_i is the coefficients estimating the effect of x_i on y and $x_1, x_2, \ldots, x_n$ are the predictor variables and ε is the error term.

Secondly, random forest employs bootstrap sampling, generating multiple subsets of the training data through random sampling with replacement. An individual decision tree receives training from each subset. Each split in a tree considers a random subset of predictors, rather than all predictors. This reduces correlation between trees, enhancing diversity and robustness in the ensemble. Thirdly, XGBoost is another tree-based model like random forest. The prime difference is that random forest builds multiple decision trees independently while XGBoost builds trees sequentially, where each new tree focuses on correcting the errors of the previous trees. Consequently, XGBoost can achieve higher accuracy than random forest while random forest is less complicated. Yet, both can handle nonlinear relationships well.

Fourthly, SVR is an extension of support vector machine. When the outcome is continuous (like temperature in this case), the SVR tries to fit as many points as possible within a margin around the hyperplane. SVR can capture nonlinear patterns in the data by using different kernels. Finally, ANNs are computational models inspired by the human brain, consisting of layers of interconnected nodes (neurons). Each neuron processes inputs, applies a weight and bias, and passes the result through an activation function. ANNs have an input layer, hidden layers, and an output layer. During training, the network adjusts weights and biases using optimization algorithms like gradient descent to minimize a loss function, which measures prediction errors. ANN is another method that excels at capturing complex and nonlinear relationships, making them effective for regression in this context.

3. **Results and Discussion.**

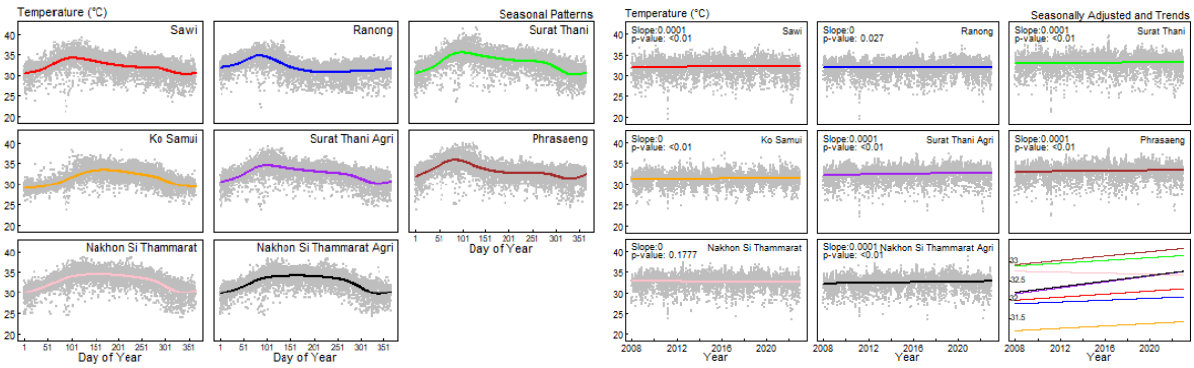
3.1. **Seasonal patterns and trends.**

Most stations exhibit a similar pattern, characterized by two temperature peaks throughout the year, as seen in Figure 3(a). The primary peak typically occurs between days 100 and 150 (April-May), while a secondary, lower peak or plateau is observed around days 200 to 250 (July-August).

All 21 stations demonstrated a warming trend (positive slope) that was statistically significant, except for Nakhon Si Thammarat, Songkhla, and Trang, where the growing trends lacked statistical significance (Figure 3(b)). The most pronounced statistically significant rising trend in daily maximum temperature, roughly 0.6°C per decade, was observed at Lanta Island and Narathiwat. A magnified view of the trends for eight selected stations is provided in the bottom-right panel of Figure 3(b). In comparison to other places, this tendency mirrored the warming trend observed in the majority of the Himalayan areas, which ranged from 0.6°C to 1.2°C per decade [23]. Nonetheless, both warming trends exceeded the pace observed in Spain, which was experiencing a warming of 0.3°C per decade [24].

3.2. **Predictive results.**

The predictive models were created by taking 4 predictors, including relative humidity, wind speed, and wind direction, to predict daily maximum temperatures at 21 stations in Southern Thailand. The best-perform model based on RMSE and R-squared of the testing set was shown in Table 2. It turned out that XGBoost outperformed other models at 13 out of 21 stations. This suggests that tree-based boosting methods capture complex relationships in the dataset well. SVR is the best model at 6 stations (Chawang, Hat Yai Airport, Phatthalung, Ranong, Surat Thani Agri, and Trang). This indicates that at these locations, the relationship between predictors and the target variable might be better captured by kernel-based methods rather than boosting techniques. While MLR performs well at two stations (Lanta Island and Narathiwat) suggesting that at these locations, the relationships between variables may be largely linear, making complex models unnecessary. However, the R-squared values (46.10% and



(a) (b)

FIGURE 3. Seasonal patterns by the natural cubic spline method of eight stations (a), and their corresponding long-term trends of each station (b)

TABLE 2. The best predictive model based on the testing by station

Station	Training		Testing		Model
	RMSE	R^2 (%)	RMSE	R^2 (%)	
1. Kho Hong Agri	1.1358	71.69	1.2537	59.55	XGBoost
2. Ko Samui	1.1002	71.06	1.3459	59.04	XGBoost
3. Nakhon Si Thammarat	1.1003	77.10	1.1806	68.08	XGBoost
4. Nakhon Si Thammarat Agri	1.1144	76.33	1.2805	66.11	XGBoost
5. Phang-Nga	0.8572	70.56	0.9532	60.97	XGBoost
6. Phrasaeng	1.0613	76.91	1.2239	67.40	XGBoost
7. Phuket	0.9167	71.96	1.1527	64.76	XGBoost
8. Phuket Airport	0.8952	64.57	1.1073	50.60	XGBoost
9. Sadao	1.0633	72.40	1.2450	62.80	XGBoost
10. Sawi	1.1096	74.17	1.4769	54.57	XGBoost
11. Songkhla	0.9341	77.58	1.0792	65.69	XGBoost
12. Surat Thani	1.174	77.29	1.3565	69.56	XGBoost
13. Yala	1.1335	76.80	1.3599	61.48	XGBoost
14. Chawang	1.4487	62.00	1.3099	68.68	SVR
15. Hat Yai Airport	1.389	58.91	1.7264	34.60	SVR
16. Phatthalung	1.2681	67.48	1.4525	47.21	SVR
17. Ranong	1.1592	71.54	1.1793	70.57	SVR
18. Surat Thani Agri	1.2755	64.96	1.4570	54.41	SVR
19. Trang	1.1751	71.79	1.3347	59.78	SVR
20. Lanta Island	1.1479	43.53	1.4170	46.10	MLR
21. Narathiwat	1.2506	58.47	1.2386	53.23	MLR

53.23%) are relatively low, indicating that these models do not capture the full variability of the data.

Consider the second-best models at each station as presented in Table 3. XGBoost is the second-best model at 6 stations (Hat Yai Airport, Lanta Island, Narathiwat, Ranong, Surat Thani Agri, Trang). While it performed well, other models (SVR and MLR) outperformed it in these cases. SVR appears as the second-best model at 12 stations. This supports the idea that SVR is a strong competitor to XGBoost, particularly when dealing with structured tabular data with some non-linearity. Moreover, some stations where

Table 3. The second-best predictive model based on the testing by station

Station	Training		Testing		Model
	RMSE	R^2 (%)	RMSE	R^2 (%)	
1. Hat Yai Airport	1.2242	68.08	1.7619	31.88	XGBoost
2. Lanta Island	0.9741	59.26	1.4345	44.53	XGBoost
3. Narathiwat	1.0369	71.40	1.2545	51.82	XGBoost
4. Ranong	1.0266	77.68	1.1899	70.04	XGBoost
5. Surat Thani Agri	1.1619	70.93	1.4768	53.16	XGBoost
6. Trang	1.0239	78.58	1.3370	59.65	XGBoost
7. Kho Hong Agri	1.2896	63.51	1.2730	58.29	SVR
8. Ko Samui	1.2537	62.43	1.3683	57.67	SVR
9. Nakhon Si Thammarat	1.2248	71.62	1.2004	67.00	SVR
10. Nakhon Si Thammarat Agri	1.2503	70.21	1.3003	65.06	SVR
11. Phang-Nga	0.9647	62.71	0.9581	60.57	SVR
12. Phrasaeng	1.1955	70.70	1.2424	66.41	SVR
13. Phuket Airport	1.0022	55.60	1.1212	49.35	SVR
14. Sadao	1.1888	65.50	1.2578	62.03	SVR
15. Sawi	1.2483	67.31	1.4840	54.13	SVR
16. Songkhla	1.0343	72.51	1.1234	62.82	SVR
17. Surat Thani	1.3260	71.03	1.3671	69.09	SVR
18. Yala	1.2503	71.77	1.3829	60.17	SVR
19. Phuket	1.1069	59.11	1.1866	62.65	Random Forest
20. Chawang	1.5934	54.12	1.3459	67.07	MLR
21. Phatthalung	1.3704	62.09	1.4883	44.82	MLR

XGBoost was the best model in Table 2, now have SVR as the second-best (e.g., Kho Hong Agri, Ko Samui, Nakhon Si Thammarat). It indicated that when the best method (often XGBoost) is not feasible or available, SVR can be a strong alternative due to its frequent appearance as the second-best method. While Random Forest is the second-best for Phuket, MLR is the second-best for Chawang and Phatthalung. ANN was not in the top two for any station suggesting that it may require a much larger dataset [25] or better tuning to outperform tree-based methods.

According to the performance of all the models, the distributions of their RMSE and R-squared were illustrated in Figure 4, left and right panels, respectively. The desirable model was supposed to have a smaller value of RMSE and a higher value of R-squared. Based on visual inspection, both the RMSE and R-squared of the testing set (Figure 4) made it clear that XGBoost, SVR, RF, and MLR perform about the same as and better than ANN. However, these performances were confirmed by Dunn's test for differences in a small subset of all possible pairs. It turned out that XGBoost and ANN are significantly better than and worse than the other three methods, respectively. On the other hand, SVR, random forest, and MLR form a middle group with no significant differences among them. In this case, the trajectories of predicted maximum temperature at Phrasaeng station in Surat Thani on the training and testing set, together with their performance measurements by MLR, RF, SVR, and XGBoost, were illustrated in Figure 5 top left, top right, bottom left, and bottom right, respectively.

4. **Conclusion.** The analysis of daily maximum temperature trends across 21 stations in Southern Thailand reveals a consistent warming pattern, with statistically significant increases at most locations. XGBoost emerged as the best predictive model at 13 stations,

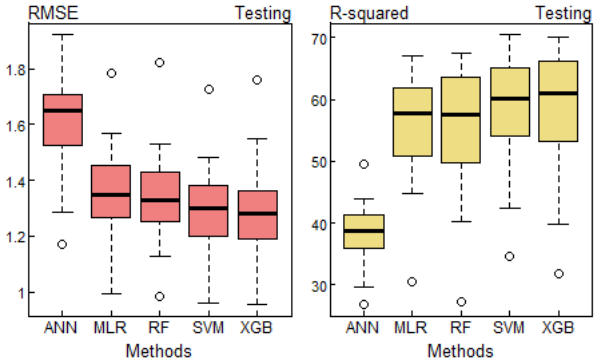


FIGURE 4. The distribution of RMSE (left) and R-squared (right) from the five models on the testing set of all 21 stations

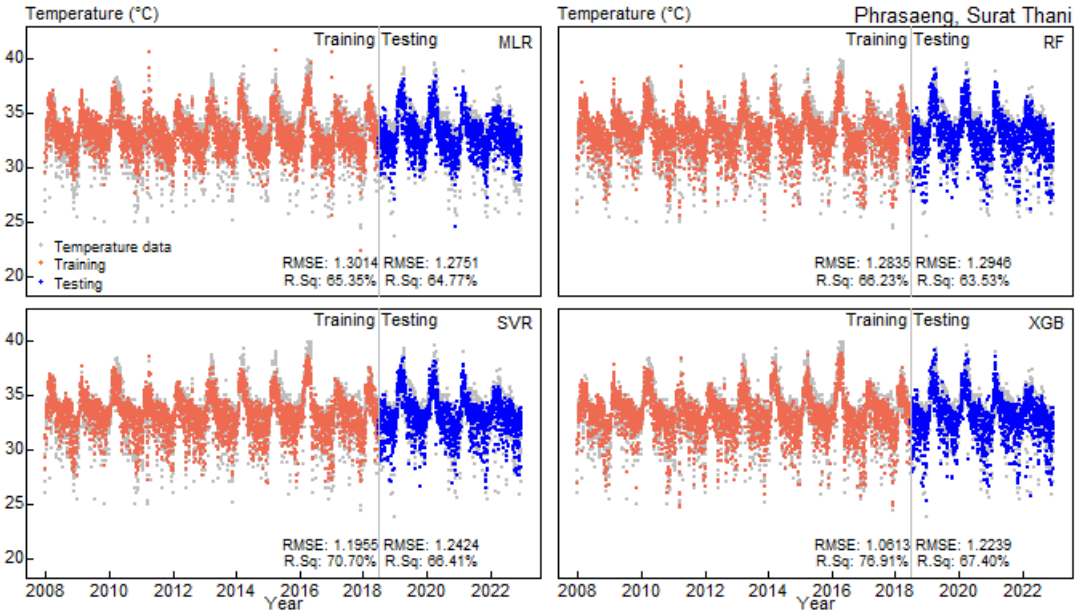


FIGURE 5. Predicted trajectories from training and testing together with their RMSE and R-squared at Phrasaeng station in Surat Thani

highlighting its strength in capturing complex relationships. SVR performed well at six stations, indicating its suitability for certain nonlinear patterns, while MLR excelled at two stations with primarily linear relationships. ANN did not rank among the top models, suggesting its limitations with the dataset size. Statistical tests confirmed that XGBoost significantly outperformed all other models, while SVR can be a strong alternative due to its frequent appearance as the second-best method.

Future studies should enhance model performance and interpretability by optimizing ANN with larger datasets and exploring explainability techniques for XGBoost and SVR. Incorporating additional variables, such as land use, vegetation index, cloud type, and air pollution, may improve predictive accuracy. Evaluating model effectiveness in forecasting extreme temperature events could aid early warning systems. Additionally, exploring deep learning architectures, as in [26], may better capture complex spatiotemporal temperature patterns.

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REFERENCES

[1] S. K. Sharma, D. P. Sharma, M. K. Sharma, K. Gaur and P. Manohar, Trend analysis of temperature and rainfall of Rajasthan, India, *Journal of Probability and Statistics*, vol.2021, no.1, 6296709, 2021.

[2] A. Chevuturi, N. P. Klingaman, A. G. Turner and S. Hannah, Projected changes in the Asian-Australian monsoon region in 1.5°C and 2.0°C global-warming scenarios, *Earth's Future*, vol.6, no.3, pp.339-358, 2018.

[3] X. Wang et al., Emergent constraint on crop yield response to warmer temperature from field experiments, *Nature Sustainability*, vol.3, no.11, pp.908-916, 2020.

[4] R. Guntukula, Assessing the impact of climate change on Indian agriculture: Evidence from major crop yields, *Journal of Public Affairs*, vol.20, no.1, e2040, 2020.

[5] R. Thompson, R. Hornigold, L. Page and T. Waite, Associations between high ambient temperatures and heat waves with mental health outcomes: A systematic review, *Public Health*, vol.161, pp.171-191, 2018.

[6] P. E. Morefield, N. Fann, A. Grambsch, W. Raich and C. P. Weaver, Heat-related health impacts under scenarios of climate and population change, *International Journal of Environmental Research and Public Health*, vol.15, no.11, 2438, 2018.

[7] I. Cohen, S. I. Zandalinas, C. Huck, F. B. Fritschi and R. Mittler, Meta-analysis of drought and heat stress combination impact on crop yield and yield components, *Physiologia Plantarum*, vol.171, no.1, pp.66-76, 2021.

[8] P. Yao, L. Qian, Z. Wang, H. Meng and X. Ju, Assessing drought, flood, and high temperature disasters during sugarcane growth stages in Southern China, *Agriculture*, vol.12, no.12, 2117, 2022.

[9] X. S. Dong, G. H. West, A. Holloway-Beth, X. Wang and R. K. Sokas, Heat-related deaths among construction workers in the United States, *American Journal of Industrial Medicine*, vol.62, no.12, pp.1047-1057, 2019.

[10] M. M. Sugg, C. E. Konrad and C. M. Fuhrmann, Relationships between maximum temperature and heat-related illness across North Carolina, USA, *International Journal of Biometeorology*, vol.60, pp.663-675, 2016.

[11] C. Yoo, J. Im, S. Park and L. J. Quackenbush, Estimation of daily maximum and minimum air temperatures in urban landscapes using MODIS time series satellite data, *ISPRS Journal of Photogrammetry and Remote Sensing*, vol.137, pp.149-162, 2018.

[12] S. Ullah et al., Observed changes in maximum and minimum temperatures over China-Pakistan economic corridor during 1980-2016, *Atmospheric Research*, pp.37-51, 2019.

[13] A. M. Abdullah, M. Ismail, F. S. Yuen, S. Abdullah and R. E. Elhadi, The relationship between daily maximum temperature and daily maximum ground level ozone concentration, *Polish Journal of Environmental Studies*, vol.26, no.3, 2017.

[14] S. M. Herath and R. Sarukkalige, Evaluation of empirical relationships between extreme rainfall and daily maximum temperature in Australia, *Journal of Hydrology*, vol.556, pp.1171-1181, 2018.

[15] E. F. Nezhad, G. F. Ghalhari and F. Bayatani, Forecasting maximum seasonal temperature using artificial neural networks "Tehran case study", *Asia-Pacific Journal of Atmospheric Sciences*, vol.55, pp.145-153, 2019.

[16] S. L. Harlan et al., Heat-related deaths in hot cities: Estimates of human tolerance to high temperature thresholds, *International Journal of Environmental Research and Public Health*, vol.11, no.3, pp.3304-3326, 2014.

[17] P. Senapeng, T. Prahadchai, P. Guayjarernpanishk, J.-S. Park and P. Busababodhin, Spatial modeling of extreme temperature in northeast Thailand, *Atmosphere*, vol.13, no.4, 589, 2022.

[18] M. Rodchuen, S. Chongcharoen and W. Bunyatisai, Trend and pattern in average monthly maximum temperatures in Thailand from 1986 to 2015, *Am. J. Appl. Sci.*, vol.17, no.1, pp.20-35, 2020.

[19] S. Worapoj, N. Nanthachai and P. Pornprom, Statistical models for daily temperature variations in Bangkok: A time series approach, *International Journal of Advances in Applied Mathematics and Computer Science*, vol.10, no.3, pp.12-18, 2023.

[20] W. Wanishsakpong and B. E. Owusu, Optimal time series model for forecasting monthly temperature in the southwestern region of Thailand, *Modeling Earth Systems and Environment*, vol.6, pp.525-532, 2020.

[21] P. Ruthirako, R. Darnsawasdi and W. Chatupote, Intensity and pattern of land surface temperature in Hat Yai City, Thailand, *Walailak Journal of Science and Technology (WJST)*, vol.12, no.1, pp.83-94, 2015.

[22] L. Beule, J.-F. Lanhenke and S. Tantanee, Trends in temperature in Thailand from 1964 to 2013, *Asia Pac. J. Sci. Technol.*, vol.21, no.4, pp.6-10, 2016.

[23] A. B. Shrestha, C. P. Wake, P. A. Mayewski and J. E. Dibb, Maximum temperature trends in the Himalaya and its vicinity: An analysis based on temperature records from Nepal for the period 1971-94, *Journal of Climate*, vol.12, no.9, pp.2775-2786, 1999.

[24] S. del Río, A. Cano-Ortiz, L. Herrero and A. Penas, Recent trends in mean maximum and minimum air temperatures over Spain (1961-2006), *Theoretical and Applied Climatology*, vol.109, pp.605-626, 2012.

[25] M. Defernez and E. K. Kemsley, Avoiding overfitting in the analysis of high-dimensional data with artificial neural networks (ANNs), *Analyst*, vol.124, no.11, pp.1675-1681, 1999.

[26] D. Andres and S. M. Isa, A comparison of three deep learning methods for predicting land surface temperature using Landsat 8 imagery, *ICIC Express Letters*, vol.17, no.11, pp.1177-1186, 2023.